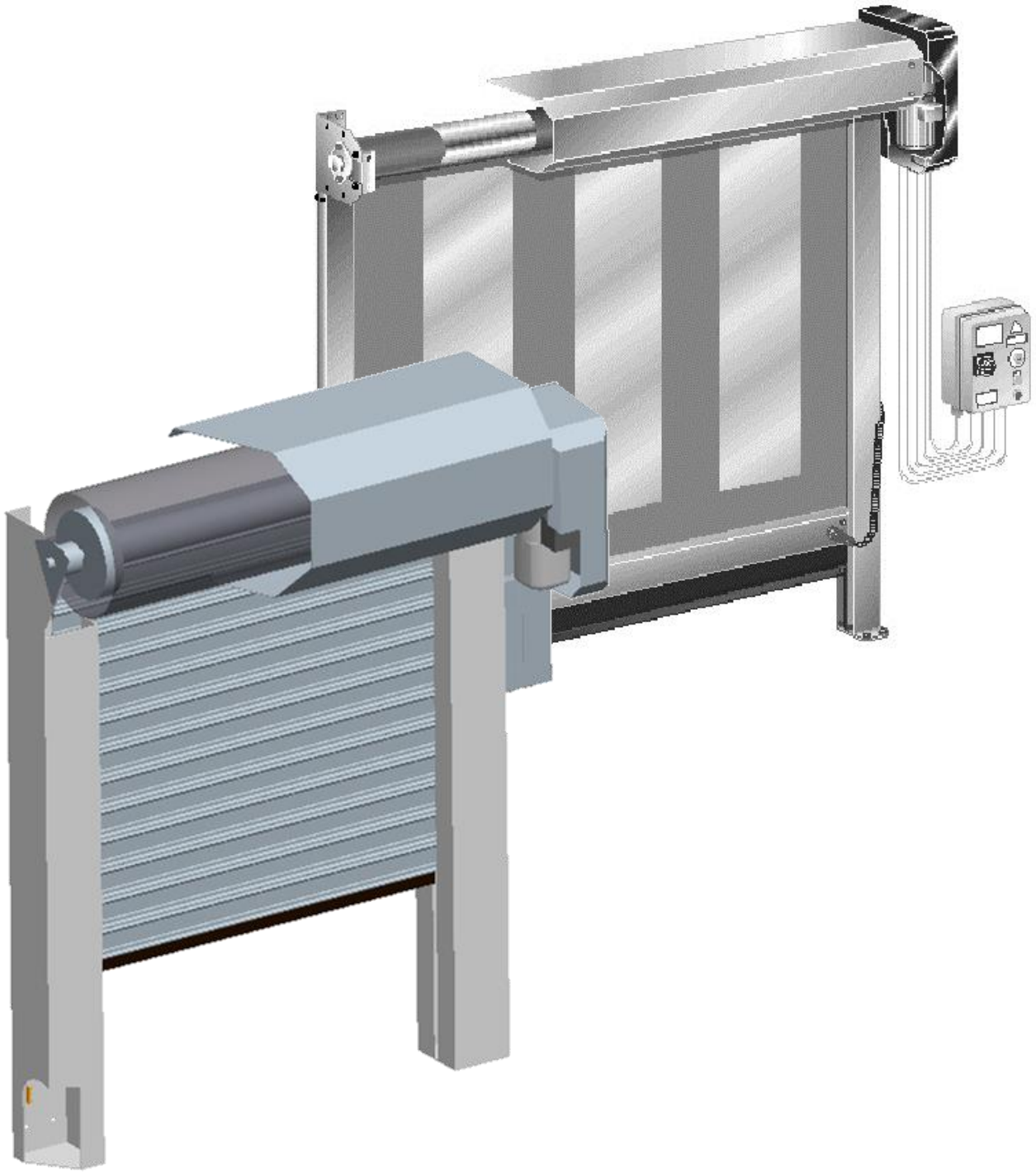
Operating Instructions

ASSA ABLOY

Albany High-Performance Doors



6410R0231

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These operating instructions are an integral part of the door. The original version of the operating instructions is in German.

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ASSA ABLOY

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1 Introduction

These operating instructions apply to the use of the following doors with associated control units: RR200, RR300, RR300PLUS, RR300 Clean, RR300 Food, RR355, RR392, RR450, RR500, RR600, RR3000, -R, -Vision, -XXL, -ISO. They constitute part of the product and describe how to use the product properly and safely throughout the product life cycle.

1.1 Applicable documents

The operating instructions are only valid in combination with the operating instructions for the control unit and the inspection book. Both of these documents are supplied with the product. A supplement to the operating instructions, which is applicable in addition to the general operating instructions, is provided for the uninterruptible power supply (UPS).

1.2 Declaration of conformity

The declaration of conformity document is enclosed in the log book which is delivered with the door.

Modifications that affect the technical data specified in the operating instructions or proper use, i.e. modifications that materially alter the machine, invalidate the manufacturer’s declaration of conformity.

1.3 Amendments and validity

The information contained in these instructions constitutes the technical specifications applicable at time of publication. Significant changes will be incorporated in a new edition of the operating instructions. The document and version number of these instructions can be found in the footer.

1.4 Document storage

Keep these operating instructions in the storage bag inside the switchbox or, for doors with an MCC control unit, in the supplied document box.

1.5 Target group

These operating instructions are intended for trained, qualified and authorised staff. All configuration activities, and in particular repairs and maintenance, may only be performed by employees of the manufacturer or other trained specialists.

1.6 Conventions

This section explains the conventions used in these instructions for safety information and operations.

1.6.1 Warnings

Warning notices give advice of existing danger for life and health of persons as well as for machine, material or environment.

The advice assembled directly at the control box and the door must be noticed and kept in readable condition at all times.

Warning notices in this document are formatted as follows:

SIGNAL WORD



Danger!

Consequences

⇨ Precautions

If the danger is an electrical hazard, the appropriate pictogram is used:

SIGNAL WORD



Danger!

Consequences

⇨ Precautions

The following warning levels are used:

SIGNAL WORD	***Hazard level***	***Consequences of non-observance***
Danger	Immediate grave danger	Death or serious physical injury
WARNING	Potential immediate grave danger	Death or serious physical injury
CAUTION	Potential hazardous situation	Minor physical injuries, material damage

This symbol leads to information explaining the various door operations.



Important information

1.6.2 Operation concept

Aim of the operation

1) Step

→ Result of the step (optional)

2) Step ...

Result of the step (optional)

2 Safety

2.1 General safety advice

These operating instructions contain important information for commissioning, operating, transporting, storing and maintaining the door safely and properly.

- Please read the operating manual carefully and keep it at a safe place for further use.
- Take particular note of the information in this section. Also note the specific safety information included in the other sections.
- Use the door only for the intended purpose. Upon transfer of this door to third parties this user manual must also be transferred to this party.

2.2 Personal protection

- No special personal protection equipment is required for using the door. Nevertheless, the door may only be used by trained and authorised staff. Furthermore, the regulations of the plant or facility where the door is used also apply.
- The use of the emergency lever (if existent) may lead to a partly self-opening or closing of the door.
- Upon failure of the electrical drive the door leaf may be unlocked and moved upwards by use of the emergency lever. At doors with emergency hand crank, the door can be opened and closed by turning the emergency hand crank which must be inserted on the bottom of the motor. This is also possible for doors with manual hand chain. In this case the drive is switched powerless.

WARNING



The door may partially open or close by itself after the emergency manual lever has been activated!

There is the risk of injuries by the movement of the door curtain.

⇨ Observe the door during use of the emergency manual lever permanently.

WARNING



Risk of injury

It can lead to injuries by simultaneous starting of the motor during inserting the emergency hand crank.

⇨ Always switch off the main switch before using the emergency hand crank.

- Do not put your hands on the side columns of the door or in the area of the top roll during operation.
- Lingering of persons in the working area of the door should be avoided. For pedestrian traffic existing wicket doors should be used.
- Go upright with normal walking speed.
- Untidiness may cause accidents. Keep the immediate area of the door tidy.
- Do not climb up the door. Do not put up a ladder against the curtain.

- When performing any type of work on the door, set the main switch (power switch) to “Off” or “0” or disconnect the CEE plug, and secure the switch or plug against accidental re-activation or reconnection. Also observe the instructions in Section 2.5.2.4, “Uninterruptible power supply (UPS)”.
- Stop using the door immediately in the event of mechanical or electrical damage, especially if the power cable or control cables are damaged.
- The door must only be operated on defined voltage or net frequency.
- Use only equipment or additional features which are authorized by the manufacturer of the door.
- Do not operate the door when excessive wind load is present.



Wind class specifications always refer to the closed position of the door.

2.3 Duty of care by the operator

The door has been designed and manufactured under consideration of a risk analysis according careful selection of valid norms and further technical specifications. The system meets status of technique and ensures the highest level of safety.

But this safety standard can only be reached in practice if all necessary measures and precautions are observed. It is the duty of the door operator to plan the respective measures and check it’s performance.

Observe the following instructions on safe operation of the door:

- That installation, initial operation, inspection, maintenance, repair work and dismantling is made only by manufacturer personnel or manufacturer trained/qualified personnel.
- That only sufficiently trained/ qualified and authorized personnel operates the door system,
- Use the machine only for its intended purpose.
- That the machine is operated only in perfectly functioning state and that the safety features are regularly checked for orderly operation,
- The operation manual (in a readable condition at all times) is kept at the site of the door system for further use if necessary,
- That the operating personnel is continuously trained with regard to working safety and operating instructions and especially the herein containing safety instructions,
- The advice assembled directly at the control box and the door must be noticed and kept in readable condition at all times.

2.4 Intended use

If not explicitly otherwise described, our doors are developed and tested for regular conditions. We would be pleased to advise you on the choice of suitable options for use under special conditions, such as persistent exposure on one side of the door to high or low temperature, positive or negative pressure, or other special ambient conditions.

The door is defined for the closing of door openings, especially on industrial buildings which are planned mostly for vehicle traffic of various types.

Exterior doors are resistant to wind and rain loads. Exterior doors can of course also be used as interior doors.

For the door RR300 Food take note of appendix D.

2.5 Safety devices

2.5.1 Electric safety devices

As electrical safety system a pre-running safety photocell, electrical safety edge or a light curtain in the door line can be used, which are complemented by a door line photocell in the side frame. All electrical safety systems are designed according to the product standard EN 13241-1. Before each closing movement of the door an internal test of the safety system and the door line photocell is being performed. If a fault is detected, the closing movement is disabled and an error message is displayed by the control system.

Doors in combination with MCC fulfill according to EN 13849-1 for the safety systems category 2 with performance level d and the emergency stop category 3 with performance level d. Doors in combination with ACS50 or MCS fulfill the safety systems category 2 with performance level c and the emergency stop category 1 with performance level c.

2.5.2 Additional protection systems

2.5.2.1 Sensors

Depending on the kind of door application it may be advisable to utilize additional sensors for protection reasons. This is valid for instance for applications involving a high frequency of human traffic as well as with low dimensioned doors less than 3 m in height and therefore no optical awareness given for the closing door as an additional protection.

ASSA ABLOY Entrance Systems offers a big variety of further protection systems for high-speed doors. light curtain, motion sensing detectors, floor loops, etc. For individual cases we suggest consultation with an Albany Door Systems representative at the application site.



At doors with a height of H ≤ 2,5m, a top roll cover is absolutely necessary.
With the RR600, the top roll cover is specified as mandatory for heights less than or equal to 2.3 m.

2.5.2.2 Emergency manual lever (except RR200, RR300, RR500, RR300 Clean and opt. RR300 Food)

At power loss, repair and maintenance work may become necessary to open the door manually:

1. By using the emergency manual lever on the side frame the brake of the drive motor can be released via a disengagement cable.

   → Depending on the counterweight set-up and the door position the door blade may partially open or close by itself.

2. If necessary, open the door further by manually pushing it upwards.

→ In the case of roll doors the door curtain will wind onto the top roll when you do this.

3. In order to reach regular function of the door again the emergency lever must be put into its original position again.
4. To use doors with ACS50 regular again, the power to the control must be turned off and on again (main switch or mains plug). For doors with MCC follow instructions on display. Once a reference search has finished the door is ready for use again.

2.5.2.3 Power on brake (Standard for RR300 Plus F+R, option for RR300 Plus)

Switching off the main switch or unplugging the power cable causes automatic door motion. If the door is closed, it opens partially, and if it is open it closes partially. The RR300 Plus R+R automatically opens by at least 2 m due to the spring-driven counterbalance.

It is not possible to operate the door with an emergency lever.



The door is not designed for a regular manual operation.

2.5.2.4 Uninterruptible power supply (UPS)

Doors with frequency converter control (only MCC) can optionally be equipped with an uninterruptible power supply (UPS). In case of a power grid failure the door can be run several times immediately after the failure. Alternativly one time up to 4 hours with the UPS in idle mode. (Precondition: the batteries have to provide the specified capacitiy)

WARNING

Risk of electrical shock



Voltage is still present on the door after mains power has been switched off.

⇨ Switch off the power switch of the UPS as well.

⇨ For working inside the UPS, the internal switch must be switched off (observe the operating instructions for the UPS).

WARNING

Risk of personal injury and material damage



Unintentional door motion can lead to damage to the product or severe personal injury.

⇨ Switch off the power switch of the UPS as well.

⇨ For working inside the UPS, the internal switch must be switched off (observe the operating instructions for the UPS).

2.6 Safety devices for RR300 Plus F+R

2.6.1 Radar motion detector

The door has a direction-detecting radar motion detector with approval for exit and escape routes. As a result the door opens automatically if persons move towards the door in the exit direction. The exit direction is defined by the installation of the radar motion detector.

If the control detects a fault in the radar motion detector the door opens automatically and remains open. It can only be put back into service after it has been repaired by a service technician.

The radar sensor must be set so that it covers the full door width and at least detects persons who are 1.5 m in front of the door. Please refer to the installation instructions of the radar motion detector manufacturer for details of how to set and adjust the detector. Reference values are in a table in chapter E 1.1 .

WARNING

Risk of personal injury and material damage

The function for exit and escape routes is not ensured, if more than one radar sensor is installed.

⇨ Connection in series of additional radar sensors is not allowed.

WARNING

Risk of personal injury and material damage

Unintentional door motion can lead to damage to the product or severe personal injury.

The function of the radar motion detector is not deactivated by the Stop-button and continues to have priority. The door is opened if a person is detected by the radar sensor, regardless of whether the Stop button is pressed or not.

⇨ The door must therefore be disconnected from the power supply before any maintenance and/or installation work is carried out.

2.6.2 Belt and spring break switch

Proper function of the counterbalance springs and belts is monitored by switches which are located in both side frames. In faultless condition the springs, belts and switches are tensioned.

If a spring or belt fails the switch is released. The door then opens electrically and switches to fault. It can only be put back into service after it has been repaired by a service technician.

The switches are to be installed by means of hanging them in the S-hooks in the hooks of the springs and fixing them to the spring mounting (Chapter E 2).

2.6.3 Catching device with solenoid actuators

The door has catching device which ensures an opening height of 2 m in the event of a fault occurring and simultaneous power failure.

This function is provided by a solenoid actuator installed at a height of 2 m in the side frame, on the drive side. If a voltage is applied to this switch, the traverse path for the

slider of the bottom profile is released; if the voltage fails the solenoid actuator locks the traverse path in the Close direction.

In normal operation of the door a voltage is applied to the solenoid actuators and the traverse path for the slider is released. If a fault is detected, e.g. a spring breaks, the door is electrically moved to the top limit position and switches to fault. If a fault occurs in this situation the curtain unwinds and the bottom profile moves downwards. However, if the power supply fails, the solenoid actuator blocks the traverse path of the bottom profile and the bottom profile is stopped at the height of the solenoid actuators and an opening height of 2m remains.

2.6.4 Regular self-tests of the door

To ensure the exit and escape route functions, self-tests are performed after switching on, after the escape route deactivation has ended (see Chap. 2.6.5), after 24 hours and after a power failure.

8:00 am is preset in the factory as the time for performance of the self-tests. This time can be changed to the user's needs using the display unit.

When the self-tests are performed, all safety relevant electrical and mechanical functions are checked. During the performance of the self-tests the exit and escape route functions are deactivated.

The electrical functions check is performed on the additional protective circuit, the control inputs, the radar motion detector and the spring and belt break switches. Further, the solenoid actuator is also actuated (see Chap. 2.6.3). The check of the mechanical functions concerns the self-opening height. This describes the achievable opening height by means of the spring mechanism, without help of the motor (e.g. in the event of a power failure). The required self-opening height is checked by releasing the brake and is 2m for use of the door in exit and escape routes. This opening height must be reached within 3s time.

The self-tests are first performed with the door closed. The start of the self-test is indicated at the display. The electrical additional functions are checked within the first 10s after starting the self-test. Then the brake is released and the door is opened by the spring mechanism (self-opening). If the opening height required for use in exit and escape routes is reached within 3s, the door stops, stops briefly and then opens up to the top limit position. Following subsequent closing of the door the self-test is completed.

If an error is found during the checking of the electronic or mechanical functions the door moves to the top limit position and switches to fault. This is indicated by an error message on the display. This error message can be acknowledged and the door can be released by input of a code by an authorised person. The self-tests are then repeated. If these tests are successful the door is ready for use again.

The allowable time interval of 10s for performance of the tests on the electronic functions is controlled by a timer relay. This relay is located in the control housing and works with dropout (OFF) delay.

CAUTION

Risk of personal injury and material damage

The door stays open if the self-test is not successful after repeated executions.

⇨ The door must be inspected by a service technician.

2.6.5 Escape route deactivation switch (optional)

If the function of the radar motion detector, the checking of the belt and spring break switches and the regular self-tests is to be suppressed (e.g. at the weekend or during company holidays), the exit and escape route functions must be deactivated by an authorised person using a key switch with the door closed. This person is responsible for ensuring that the exit and escape route functions are activated if the building is being used by people again, i.e. is occupied.

A display at the operator unit of the MCC control provides information about the condition of the exit and escape route deactivation.

In the event of a power failure the door opens automatically and remains in the open position, even when the power failure is corrected. The door can therefore not be viewed as being burglary or theft protection if the exit and escape route functions are deactivated.

WARNING

Risk of personal injury and material damage



The function for exit and escape routes is not ensured, if the escape route deactivation switch is used.

The door remains closed and it can then not be used either as a door or as an escape route.

⇨ The exit and escape route functions must be activated if the building is being used by people again, i.e. is occupied.

2.7 Residual risks

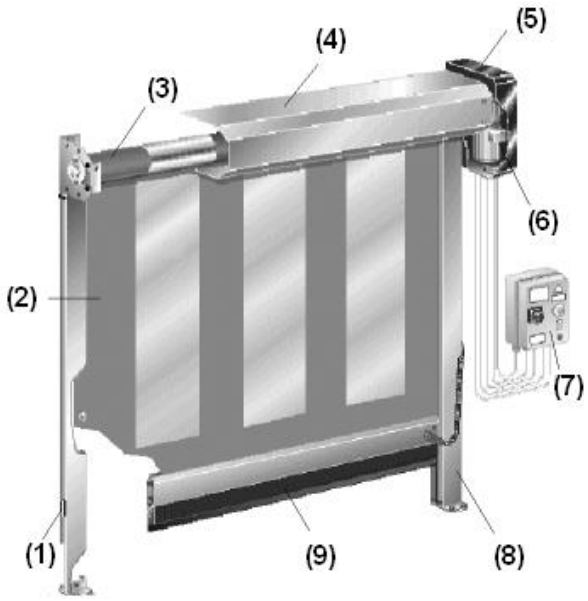
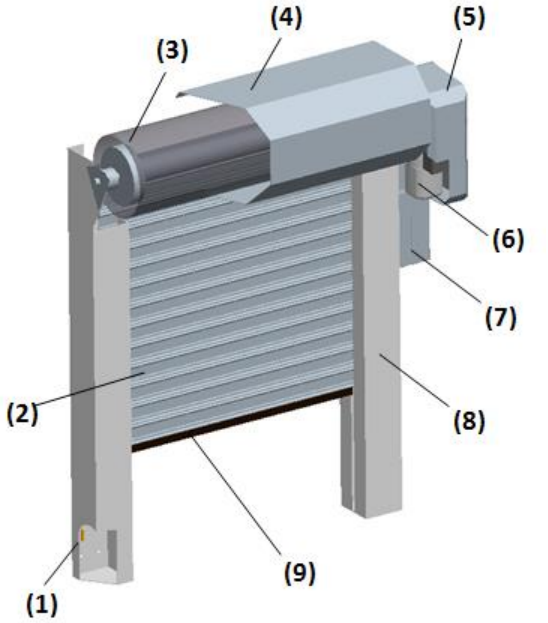
The door has been designed and manufactured under consideration of a risk analysis according careful selection of valid norms and further technical specifications. The system meets status of technique and ensures the highest level of safety. But this safety standard can only be reached in practice if all necessary measures and precautions are observed. In particular, when the door is used by people, it must be ensured that they have been instructed in the use of the door and advised of the potential danger of walking under the door while it is closing.

3 Door Product Description

The operation and construction of the door are described below. Consult the appropriate tables to determine the configuration options available for specific door types.

3.1 Door components

The figure shows the Albany RR300 and Albany RP 2000 doors. Other types of door may differ in detail.

Doors with flexible curtains	Doors with rigid curtains
	

1	Photocell, light curtain	6	Drive system
2	Door curtain	7	Electric control box
3	top roll	8	side frame
4	Option: Top roll cover (Aluminum anodized, steel galvanized, stainless steel)	9	Continuous safeguarding, electrical safety edge, rubber profile
5	Option: Motor cover (Plastic, stainless steel, steel powder coated, aluminium)		

The door line photocell or light curtain (1) safeguards the closing area of the door. Disruption of the beam prevents or stops the closing movement and the direction of travel is reversed.

Depending on the type, flexible door curtains **(2)** are made from coated, dyed and woven polyester monofilament or from transparent or coloured PVC with draught strips. Rigid curtains are made from aluminium slats.

The top roll **(3)** has a tubular aluminium shaft with glued-in shaft stubs or a steel shaft. Doors with rigid curtains have a special DiscDrive system that prevents direct contact with the curtain surfaces during roll-up.

The drive unit **(6)** contains an electric motor, gearbox and limit switches.

The electrical switchbox **(7)** contains an ACS50, MCS or MCC control unit, depending on the configuration.

The side frames **(8)** are made from galvanised steel, aluminium or stainless steel and may also contain a counterbalance system, depending on the door model.

Upon touching the safety edge (9) the door reverses into the open position immediately. A pre-running photocell may be used as an alternative.

3.2 Operation and installation of the door systems

Albany doors are vertically opening high-speed roll-up doors. They consist of two side frames, a top roll with an electric drive unit, a door curtain and a bottom profile.

The side frames guide the bottom profile and the door curtain, which is rolled onto the top roll when the door opens.

During closing of the door the curtain rolls down from the top roll and is being pulled down by the self weight of the bottom profile and the door curtain. In the case of doors with built-in curtain tensioning, the door curtain is also actively tensioned by the force of the curtain tensioning system.

Doors with a counterbalance system have this system housed in the side frames. The counterbalance assists the drive unit during door opening and also enables manual opening, for example in case of power failure, by causing the door to open partially on its own.

The doors in the Albany RR450 series also have optional travelling wind bars, which can be fitted either on the front or the rear of the door as necessary.

Type	***Curtain tension***	***Counterbalance***	***Side frame covers***
RR300	No	No	None
RR300 Plus	optional	Springs	screwed on
RR300 Clean, RR200, RR500	No	No	screwed on
RR300 Food	No	weights (optional)	hinged
RR355	Yes	Springs	screwed on
RR392	Yes	weights	screwed on
RR450, RR600	Yes	Springs	hinged
RR3000, -R, -Vision	Yes	Springs	screwed on
RR3000 ISO, -XXL	No	Springs	screwed on

3.2.1 Safety systems

All RapidRoll doors are fitted with standard-compliant protection on the main closing edge.

In addition, damage to the door in the event of a collision can largely be avoided by an (optional) breakaway mechanism.

The following systems are available:

Breakaway mechanism	***Function***
Manual	After a collision the crash system can be manually repaired.
Self-repairing	The breakaway mechanism is restored to the operating position by pressing a button.
SRD	Automatic breakaway mechanism minimises collision damage

The safety systems fitted to the doors are listed in the following table.

Type	***Securing the main closing edge***	***Breakaway mechanism *)***
RR300, RR300 Plus	electrical safety edge	Manual
RR200, RR500	electrical safety edge	Self-repairing
RR300 Clean	electrical safety edge or optional pre-running photocell	None
RR300 Food	electrical safety edge or optional pre-running photocell	Manual
RR355	pre-running photocell or optional electrical safety edge	Self-repairing
RR392	electrical safety edge	Manual
RR450	pre-running photocell or optional electrical safety edge	Self-repairing
RR600	pre-running photocell or optional electrical safety edge	SRD or Self-repairing
RR3000, -R, -Vision, -XXL	pre-running photocell or optional electrical safety edge	None
RR3000 ISO	Light curtain up to 2,5m height	None

* The type of breakaway mechanism depends on the type of bottom edge protection

All doors also have a door line photocell system integrated in the side frames. Disruption of the door line photocell or triggering the safety edge leads to a complete stop of the door curtain. After the door curtain stops it will reverse to the open position.

3.2.2 Slats (only doors with rigid curtains)

The slats are individually suspended, double-walled extrusions sealed relative to each other.

Type	***Slats***	***Window(s) ****
RR3000, -R, -XXL	Double-wall anodised	optional
RR3000 Vision	Window frames anodised with single glazing	Full
RR3000 ISO	foamed sandwich-panel with steel sheets RAL 9006	optional

* The number depends on the door size and type, among other things

3.3 Emergency exit door Albany RR300 plus F+R

3.3.1 Conditions of Use

These instructions are directed at the owner/operator of the Albany RR300 Plus F+R emergency exit door. The owner/operator is responsible for correct use, instructing the operating personnel and for technical maintenance of the door.

These instructions for the basis for intended use and faultless functioning of the door.

The Albany RR300 Plus F+R has been built to state of the art standards and the recognised safety guidelines for exit and escape routes and has been approved by an independent test body.

Any change whatsoever to the door results in withdrawal of the operating permit as an emergency exit door; the manufacturer's liability is cancelled by any failure to observe this requirement or in case of deliberate misuse.

The door system may be disconnected from the mains in emergencies only. Disconnecting it over night is not allowed!

3.3.2 General information

The Albany RR300 Plus F+R conforms to the essential requirements of the “Richtlinie über automatische Schiebetüren in Rettungswegen” (AutSchR - Guidelines for Automatic Sliding Doors in Escape Routes). The consent of the responsible building authority must always be obtained for approval for use of the door in exit and escape routes.

The door has an automatic opening function to make it suitable for use in exit and escape routes. This ensures opening of the door as soon as persons move towards the door in the exit direction.

The door has various monitoring and control devices to ensure the exit function even in the event of a power failure or damage (spring wear, spring break, belt break). The exit and escape route function exists if the opening height is at least 2m.

The door is delivered with a display on which information for operation and about the status of the door is displayed. It is therefore recommended that this display be installed in the immediate vicinity of the door, if the display is not planned to be mounted on the side frame.

3.4 Control unit

Albany doors may be operated with a powerful frequency converter control unit MCC respectively MCS-FC, which enables soft acceleration and deceleration of the door. Alternatively, several types of doors can be operated with a contactor control unit ACS50 respectively MCS-R. See the table in the appendix (Section A 3) for the maximum closing speeds of the individual door types.

DANGER!



Risk of electrical shock

Operating doors with drive units or control units other than the recommended types may create a risk of electrical shock.

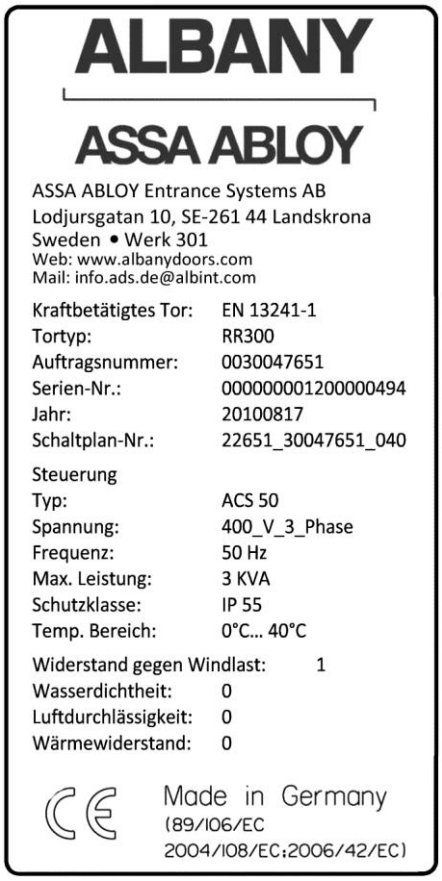
⇨ The door must only be operated with an original drive and control unit from the manufacturer.

3.5 Drive

The high-speed doors are operated by electrical drives. The drive unit is installed at the upper part of the door, vibration muffled and not accessible from the floor. The position of the drive unit may be selected by the customer for the right or left hand side. The temperature of the drive unit is safeguarded by a thermo element, which disrupts the power supply of the drive upon overheating.

In the case of doors without counterbalance, the gearbox is equipped with a catching device.

3.6 Identification plate



(the actual data will vary depending on the specific door)

3.7 Technical Data

The inspection book contains the following technical data:

Technical Data	***Location***
Door dimensions	Principle drawing
Dimensions of the installed door	
Fitting clearances	
Fixing points	Fixing plan

See the appendix (Sections A 1 and A 2) for all other technical data for the door.

ASSA ABLOY Entrance Systems reserves the right to make changes to technical information listed in this document.

4 Operation

This section describes the operation of the door, irrespective of the control unit used. See the control unit user manual for a description of the control unit concerned.

4.1 Safety

- Do not use this door for purposes it is not intended for.
- Upon damage of the door (mechanical or electrical) it must immediately be set off operation. This is especially valid on damages of the power net or control cables.
- Use only equipment or additional features which are authorized by the manufacturer of the door.
- Observe the safety instructions in Section 2 also.

4.2 Operation

The door has an electrical drive unit and can be operated using the buttons on the door control unit or by additional external actuators and safety devices.

The Open, Stop and Close buttons on the door control unit should normally be used during door setup, maintenance and repair activities.

In regular operation, the door is controlled by external signal sources such as pull-cord switches, induction loops, radar or radio detectors or mushroom-head buttons connected to the door control unit via dry contacts.

4.3 Malfunction

DANGER



Risk of personal injury and material damage

Improper troubleshooting can lead to damage to the product and serious personal injuries.

⇨ Troubleshooting must always be performed by suitably qualified staff.

⇨ Switch off power to the door and secure it against accidental re-activation. Also observe the instructions in Section 2.5.2.4, “Uninterruptible power supply (UPS)”.

⇨ Secure the main switch or CEE connector against being switched on again by locking it or by having a second person present.

⇨ Secure the action area of the door.



If the error cannot be repaired please contact the after market department of the supplier or manufacturer of the door.

4.3.1 Mechanical malfunctions

Malfunction	*Possible causes*	*Necessary measures*
Door opens independently	Brake at the drive defect	Exchange the brake
	Brake wrongly adjusted	Adjust the brake
	Counter-balance is oversized	Call after market
Manual opening not possible	Counter-balance defective	Exchange the counter-balance devices
	Bowden cable hanging off or too loose	Check correct position and tension of the bowden cable
	Brake blocking	Call after market
Drive without any function	Power supply is off	Connect the door to the power
	Electrical connections defective	Check electrical connections
	Fuse has blown	Replace fuse
curtain unwraps inclined	Door not correctly installed	Call after market
	An object was rolled up	Close the door and check for enclosed object

4.3.2 Electrical faults

Malfunction	*Possible causes*	*Necessary measures*
Door without any function	Power supply is off	Check power supply, net fuse
	Control system fuse blown	Check control unit and exchange the fuse
	Main switch off	Check (main switch) switch on
	Stop-circuit interrupted	Check stop circuit. Not used stop inputs must be bridged in accordance to the circuit diagram
Door does not close	Up impulse permanently activated	Test impulse actuator
	Photocell interrupted / disconnected	Test safety devices
Door close only with next impulse	Toggle circuit adjusted	Check adjusted control function

5 Installation

DANGER



Risk of injury

Improper handling of the packaged door can lead to injury.

⇨ The palette must be placed on even ground.

⇨ Do not remove the pallet before the start of installation.

5.1 Inspecting the delivery

The door is mainly preassembled when delivered. The side columns are fixed together with the barrel cover, the motor cover (if ordered), fixing material and control unit onto the transport device (wooden palette). Check the door for damage during transport.

- Check the delivered items against the order documents to ensure that the delivery is complete.

5.2 Fitting preparations

DANGER



Risk of injury

Incorrect installation of the door can lead to injury.

⇨ The door must only be installed by trained/authorized personnel.

⇨ The fitting should only be done by manufacturer personnel/subcontractor or manufacturer trained personnel.

⇨ Do not remove the straps securing the door curtain until after the drive unit has been fitted.

Complete the following preparations before installing the door:

- Check the fitting site with regard to necessary fixing points and lift up conditions. If necessary a supporting frame or other fixing possibilities must be made. The upper fixing points should be screwed onto a steel supporting frame or concrete element. Consult the installation drawing in the inspection book.
- Check measures of the actual dimensions (width and height) of the door opening and compare with order papers.
- The fitting site must be secured against pedestrian and vehicle traffic.
- The plastic packing of the door must be removed before fitting.
- For the door RR300 Food take note of chapter D.

DANGER!



Risk of injury

Unsecured guiding cover can fall down for door types RR3000.
By pushing up and unhook the bottom guiding cover the upper side cover can be unmounted.

- ⇨ The door must only be installed by trained/authorized personnel.
- ⇨ The fitting should only be done by manufacturer personnel/subcontractor or manufacturer trained personnel.
- ⇨ The upper side cover must be mounted when removing the bottom guiding cover.

5.3 Installation

The fitting team has a detailed installation manual. In case of questions, contact ASSA ABLOY Entrance Systems.



The door must be connected to the potential equalisation rail (min. 6 mm²). Therefore is an earthing screw in the area of the drive unit or in the lower part of the side frame on the drive side which is marked with an earth symbol.

5.4 Dismantling

To dismantle the door, follow the installation instructions in reverse order.

6 Life Cycle

6.1 Installation

See the user guide for the associated control unit.

6.2 Maintenance and Cleaning

In case of necessary work of maintenance, malfunction, repair, please ask your supplier or manufacturer of the door.

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6.2.1 Safety instructions

The maintenance and inspection work must only be performed by manufacturer trained/ authorized personnel.

- Before performing any maintenance and cleaning work, disconnect the door from the mains by switching off the lockable main switch on the switch box or unplugging the CEE connector. Also observe the instructions in Section 2.5.2.4, “Uninterruptible power supply (UPS)”
- Use a meter or other test device, verify that the switch box is de-energised.
- Please observe that all used ladders, scaffoldings, or such, correspond with the valid safety regulations.
- The working area must be blocked for traffic. Lubricating oil on the floor, tools and other material should be removed from the floor when work has been finished.

6.2.2 Inspection plan

Record all maintenance, repairs and modifications in the inspection list included in the inspection book for every door, to ensure that all changes can be traced.

The necessary service events, inspections and implementation notes are documented in the inspection plan in chapter B.

WARNING

Risk of personal injury and material damage

RR300 Plus F+R: The function for exit and escape routes is not ensured, if maintenance plan is not followed.

⇨ Compliance with the inspection plan is necessary in order to maintain the exit and escape routes characteristics. Special points of the inspection plan for the RR300 Plus F+R have priority.

DANGER

Risk of explosion

RR450Ex / 600Ex: Improper maintenance nullifies the Declaration of Conformity with regard to the ATEX directive.

⇨ Compliance with the inspection plan is necessary in order to maintain the explosion protection characteristics. Maintenance activities and extended downtime periods (if any) should be planned in.

6.2.3 Cleaning and care

6.2.3.1 Door curtain

Pay particular attention to the cleaning instructions in the appendix for the RR300 Clean (C) and RR300 Food (D 2) doors.

CAUTION

Risk of material damage

Improper cleaning of the curtain may cause damage to the surface.

⇨ Do not use any glass cleansing liquids (aggressive contents)

⇨ Never use score cleansers, scrapers, razor blades and spatula, etc.

⇨ Clean with warm water, small dose of mild plastic cleanser and a soft non-bobble cloth.

DANGER

Risk of personal injury and material damage

Pollution of the curtain can cause reduction (loss) of the protection property for laser protection doors.

⇨ The surface facing to the laser must be kept clean.

6.2.3.2 Side Frames

CAUTION

Risk of material damage

Improper cleaning of the door frame may cause damage to the surface.

⇨ Do not use high pressure cleaning machine.

⇨ Dust may be removed with a soft cloth.

⇨ Harder dirt may be removed with water and general liquid cleanser.

⇨ Dirt of grease or oil on metal surface may be removed with solvent containing cleanser.

6.2.3.3 Surroundings

Especially after maintenance work the floor around the side columns may be dirty with lubricants, etc. Carefully clean these areas.

6.3 Decommissioning

- Follow the installation instructions for dismantling.
- Disconnect power from the door before dismantling. In particular, ensure that all springs are de-tensioned. Also observe the instructions in Section 2.5.2.4, “Uninterruptible power supply (UPS)”
- Check parts that are subject to wear and replace them if necessary before reassembly.
- Check all connectors and avoid kinking cables.
- Store all components in a clean, dry place.

6.4 Storage and transport

6.4.1 Transport

The single door components must only be removed from the packaging at the site of installation.

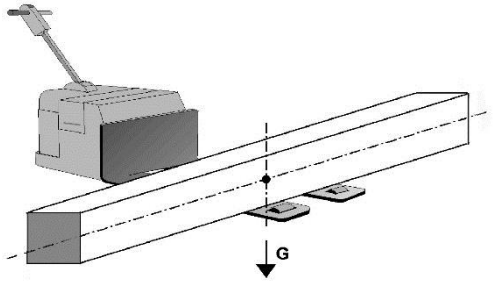
DANGER

Risk of injury

Improper transport may cause the load to slip and fall.

⇨ Observe the legal safety regulations when transporting the door to the fitting site.The door must be secured against tilting or slipping.

⇨ Lift the door at its centre of gravity to prevent it from tipping sideways.



6.4.1.1 Dimensions of the door with packaging

	Transport pallet [mm]		*Drive and control unit pallet [mm x mm]*
Door	***Length***	***Width***	***Length x width***
RR200	W + 1000	530	800 x 600
RR300, RR300 Plus, RR300 Clean	H + 500	500	800 x 600
RR355, RR300 Food	H + 500	650	800 x 600
RR392	H + 800	750	800 x 600
RR450, RR600	H + 800	750	800 x 600
RR500	W + 1000	750	800 x 600
RR3000, -R, -Vision; pallet 1 (top roll)	W + 850	750	
RR3000 XXL, -ISO (H<4700); pallet 1 (top roll)	W + 850	850	
RR3000 ISO (H>4700); pallet 1 (top roll)	W + 850	980	
RR3000, -R, -Vision, -ISO, -XXL pallet 2 (side frames)	H + 600	750	

W = width of door opening; H = height of door opening

6.4.1.2 Weight of the door with pallet [Kg±10%]

(Intermediate values can be interpolated; weights may vary depending on specific configuration options.)

W X H [m]	RR200	RR300, RR300 Clean	RR300 Plus, RR300 Food	RR355	RR392	RR450	RR500	RR600	RR3000, Vision top roll	RR3000, Vision side frame	RR3000 R top roll	RR3000 R side frame	RR3000 ISO, XXL top roll	RR3000 ISO, XXL side frame
2 x 2	116	140	175	250										
2 x 2,5		175	215		400									
3 x 3	149	210	260	300	470	400	370	480	270	330	250	270		
4 x 4	184	270	330	380	630	530	445	610	380	420	325	340	430	520
5 x 5			400		770	650	535	740	500	500			580	630
6 x 6					920	780	620	870					750	770

6.4.2 Storage

The partly preassembled door is being delivered on a transport resp. mounting palette completely wrapped with plastic material. Undamaged packing allows storage of the door outside of buildings for several days.

6.5 Disposal

The packing material of the door may be returned to the manufacturer. Defect door components should be disposed of at the site according environmental regulations.

Defect electronic components should be disposed to special waste disposal.

A. Appendix

The following tables list technical data for the doors. A capsule summary of the maximum opening and closing speeds is also provided in Section A 3.

A 1. Technical Data: Doors with flexible curtains

	Albany RR300	***Albany RR300 Plus***
Door principle:	Roll door	Roll door
Running direction:	vertical	vertical
Application:	Interior door	Interior door
Wind class:	1	1
Available sizes:		
W min:	500 mm	1000 mm
W max:	4000 mm	5000 mm
H min:	500 mm	1000 mm
H max:	4200 mm	5000 mm
Drive:	electrical	electrical
Mains power:	See control unit manual	See control unit manual
Motor performance:	0,65 kW; 0,75 kW	0,65 kW; 0,75 kW
Control unit:	MCC, MCS-FC MCS-R, ACS50 (50 Hz only, W max. 3500 mm, H max. 3500 mm)	MCC, MCS-FC MCS-R, ACS50 (50 Hz only, W max. 4000 mm, H max. 4200 mm)
Control voltage:	24 V DC	24 V DC
Operating temperature range:	+10°C - +40°C	+10°C - +40°C RR300 Plus F+R: +10°C - +30°C
Motor protection:	IP55; IP54 with MCS	IP55; IP54 with MCS
Side frame:	Galvanised steel or stainless steel	Aluminum or stainless steel
Door curtain:	PVC, Rolltex, Nomatex, Travitop	PVC, Rolltex, Nomatex, Travitop
Lifting device:		Pulley, cable and belt
Sound pressure level:	70 dB A	70 dB A

	Albany RR300 Clean	***Albany RR300 Food***
Door principle:	Roll door	Roll door
Running direction:	vertical	vertical
Application:	Interior door	Interior door
Wind class:	0	1
Available sizes:		
W min:	1000 mm	1000 mm
W max:	3500 mm	4000 mm
H min:	1000 mm	1000 mm
H max:	3500 mm	4000 mm
H max (ACS50):		3500 mm
Drive:	electrical	electrical
Mains power:	See control unit manual	See control unit manual
Motor performance:	0,75 kW	0,75 kW
Control unit:	MCC, ACS50 (50 Hz only)	MCC, ACS50 (50 Hz only)
Control voltage:	24 V DC	24 V DC
Operating temperature range:	+10°C - +40°C	0°C - +40°C
Motor protection:	IP55	IP55
Side frame:	Galvanised steel or stainless steel	stainless steel
Door curtain:	PVC	PVC, Travitop
Lifting device:		Pulley and cable
Sound pressure level:	70 dB A	70 dB A

	Albany RR200	***Albany RR500***
Door principle:	Roll door	Roll door
Running direction:	vertical	vertical
Application:	Interior door	Exterior and interior door
Wind class:	1	3 (4 if DW<6000)
Available sizes:		
W min:	1000 mm	1500 mm
W max:	4000 mm	8000 mm
H min:	1000 mm	2300 mm
H max:	4000 mm	5500 mm
Drive:	electrical	electrical
Mains power:	See control unit manual	See control unit manual
Motor performance:	0,75 kW	0,75 kW
Control unit:	MCS	MCC
Control voltage:	24 V DC	24 V DC
Operating temperature range:	+10°C - +40°C	-10°C - +40°C
Motor protection:	IP54	IP55
Side frame:	anodized aluminum	anodized aluminum
Door curtain:	PVC	PVC, Nomatex
Lifting device:		belt
Sound pressure level:	70 dB A	70 dB A

	Albany RR355	***Albany RR392***
Door principle:	Roll door	Roll door
Running direction:	vertical	vertical
Application:	Exterior and interior door	Exterior and interior door
Wind class:	2	2
Available sizes:		
W min:	1000 mm	2000 mm
W max:	4000 mm	6000 mm
H min:	1500 mm	2000 mm
H max:	4200 mm	6000 mm
Drive:	Electrical (chain drive optional)	Electrical (chain drive optional)
Mains power:	See control unit manual	See control unit manual
Motor performance:	1,1 kW	1,1 kW
Control unit:	MCC, ACS50 (50 Hz only)	MCC, ACS50 (50 Hz only)
Control voltage:	24 V DC	24 V DC
Operating temperature range:	-10°C - +40°C	-10°C - +40°C
Motor protection:	IP55	IP55
Side frame:	anodized aluminum	galvanized steel
Door curtain:	PVC, Rolltex, Nomatex	PVC, Rolltex
Lifting device:	Pulley and cable	Pulley and cable
Sound pressure level:	70 dB A	70 dB A

	Albany RR450	***Albany RR600***
Door principle:	Roll door	Roll door
Running direction:	vertical	vertical
Application:	Exterior and interior door	Exterior and interior door
Wind class:	2	3
Available sizes:		
W min:	2000 mm	1500 mm
W max:	7500 mm	6000 mm
H min:	2000 mm	2000 mm
H max:	6000 mm	6000 mm
Drive:	Electrical (chain drive optional)	electrical
Mains power:	See control unit manual	See control unit manual
Motor performance:	1,1 kW 1.5 kW (explosion-proof doors)	1,1 kW
Control unit:	MCC, ACS50 (50 Hz only)	MCC
Control voltage:	24 V DC	24 V DC
Operating temperature range:	-10°C - +40°C	-10°C - +40°C
Motor protection:	IP55	IP55
Side frame:	anodized aluminum	anodized aluminum
Door curtain:	PVC, Rolltex	RollTex, Nomatex, PVC
Lifting device:	Pulley, cable and belt	Pulley and steel cable
Sound pressure level:	70 dB A	70 dB A

A 2. Technical Data: Doors with rigid curtains

Albany RR	***3000***	***3000 Vision***	***3000 R***	***3000 ISO***	***3000 XXL***
Door principle:	Roll door				
Running direction:	vertical				
Application:	Exterior and interior door				
Wind class:	3	3	3	2 – 4	2
U-value:	-	-	-	1,5 W/m^{2}K	
Available sizes:					
W min:	600 mm				
W max:	5000 mm	5000 mm	6000 mm	7000 mm	8000 mm
H min:	3125 mm	2500 mm	1500 mm	2500 mm	2500 mm
H max: (height increment 125 mm)	5000 mm	5000 mm	3000 mm	6000 mm (height increment 100 mm)	6000 mm
Drive:	electrical				
Mains power:	See control unit manual				
Motor performance:	1,1 kW				
Control unit:	MCC				
Control voltage:	24 V DC				
Operating temperature range:	-10°C - +40°C				
Motor protection:	IP55				
Side frame:	galvanized steel				
Door curtain:	Anodized aluminum slats, some with windows RR3000 ISO: Foamed sandwich-panel with steel sheets RAL 9006, Optional with double-glazed windows				
Lifting device:	Flat belt with reinforcing steel ropes, belt				
Sound pressure level:	70 dB A				

A 3. Maximum opening and closing speeds

	MCC		***ACS50***		***MCS-FC***		***MCS-R***	
m/s (max)	Open	Close	Open	Close	Open	Close	Open	Close
RR200					1,5	0,6		
RR300	2.0	1.0	1.0	0.8	1,5	0,8	1,0	1,0
RR300 Plus	2.2	0.9	1.2	1.2	1,7	0,8	1,2	1,2
RR300 Clean	2.0	1.0	1.0	1.0				
RR300 Food	2.2	1.0	1.0	1.0				
RR392	2.2	0.9	0.8	0.8				
RR450	2.2	0.9	0.8	0.8				
RR500	1.0	0.5						
RR600	2,0 or 3,2*)	1.2						
RR3000	2.2	0.7						
RR3000 R	2.0	0.6						
RR3000 ISO	2,1	0,7						
RR3000 Vision	1.8	0.7						
RR3000 XXL	1.8	0.6						

* Depending on door size

B. Appendix Inspection

The following tables list inspection plans for the doors. The overview of the service intervals are provided in section B 1. The numbers are used in the inspection plan 0.

Information for RR300 Clean doors for GMP-relevant aspects are provided in section B 2.

B 1. Service intervals

1	half-yearly or 50.000 cycles (* only for RR300 Plus F+R)
2	annually or 50.000 cycles
3	annually or 100.000 cycles
4	annually
5	clean if necessary
6	every 4 years
7	150.000 cycles
8	250.000 cycles
9	500.000 cycles
10	150.000 cycles; RR300 Plus F+R: 100.000 cycles
11	with MCC: 150.000 cycles; with ACS50: 50.000 cycles
12	RR450Ex / 600Ex: 200.000 cycles
13	450.000 cycles; RR450Ex/600Ex 200.000 cycles
14	200.000 cycles; RR600 Crash: 100.000 cycles
15	DH < 3m: 500.000 cycles; DH ≥ 3m: 250.000 cycles
16	DH < 4m: 300.000 cycles; DH ≥ 4m: 200.000 cycles
**	Exchange of bearings only by drive manufacturer or an approved workshop for the maintenance of the explosion-proofness. (downtime at the time of external execution of the exchange, which can be expected, approx. 8-10 working days)
***	Regular conditions: central European climate, door mounted interior or exterior with covers, no aggressive ambient conditions

≤150.000 cycles: ±10.000 cycles; >150.000 cycles: ±20.000 cycles

B 2. GMP-relevant inspection (only RR300 Clean)

Could be done also by the customer.

Mechanical components	**proposal**
Check air leakage: Curtain should permit only small air leakage around gaps	3 months or 25000 cycles
Motor sealing: Motor cover should fit firm and tight, cable glands must be orderly	3 months or 25000 cycles

B 3. Inspection plan

Inspection	RR300,RR200	RR300 Plus	RR300 Clean	RR300 Food	RR355	RR392	RR450	RR500	RR600	RR3000,R,Vision	RR3000 XXL	RR3000 ISO
Installation fixings												
Tighten screws	1	1	2	1	1	1	1	1	1	3	2	3
Check welding seams	1	1	2	1	1	1	1	1	1	3	2	3
Drive												
Check the fixing of the torque arm and buffer	1	1	2	1	1	1	1	1	1	3	2	3
Check brake condition and function of break release lever	1	1	2	1	1	1	1	1	1	3	2	3
Check operation smoothness and function of the emergency manual lever, maybe lubricate		1		1	1	1	1		1	3	2	3
Check the chain, adjust and possibly retension and lubricate					1	1	1	1		3	2	2
RR450Ex / 600Ex: Change bearings in motor**:							6					
Drive shaft												
Check screws to drive shafts and flange bearings	1	1	2	1	1	1		1	1	3	2	3
Lubricate pedestal bearing and retighten it's fixation							1					
Check the drive shaft in the aluminium shaft for secure fit	1	1	2	1	1	1	1					
Check tight position of belt and rope discs		1		1	1	1	1	1				
Check the condition of the cable drums									1			
Check retainer of spiral disk on the shaft										3	2	3
Chain drive: check fastening of locking element, tighten the screws with 18 Nm										3	2	2
Check condition of sound isolation profiles at the spiral disks										3	2	2
Check flat belts for wear and damage; check connection to spiral disks										3	2	2
Door curtain												
Check curtain for cracks and traces of scrapes or shavings in the guides	1	1	2	1	1	1	1	1	1	3	2	2
Check fixation of door leaf at the top roll and at the bottom profile	1	1	2	1	1	1	1	1	1			
Check tight position and wear & tear of the wind tabs	1	1	2	1	1	1	1	1	1			
Belt: condition, lubrication, wear; if necessary replace teeth	1							1				
Check exact position of slat sealing										3	2	3
Check fixing of slats on pulling device and endcaps										3	2	2

Inspection	RR300,RR200	RR300 Plus	RR300 Clean	RR300 Food	RR355	RR392	RR450	RR500	RR600	RR3000,R,Vision	RR3000 XXL	RR3000 ISO
Check slats around adapters for damage (especially 1-5 from top)										3	2	2
Check solid position of slat adapter										3	2	3
Bottom profile												
Check tight fit and condition of profile, guiding pieces and the gliding pieces	1	1	2	1	1	1	1		1			
Check condition of the safety edge and the cable	1	1	2	1	1	1	1	1	1			
Check condition of the coily cable	1	1			1	1			1			
Check the condition and function of the crash system	1	1		1	1	1	1	1	1			
Check condition and function of wireless safety edge; change battery if necessary	7	7						7				
side frame												
Check fixing and condition of the profiles with attachments	1	1	2	1	1	1	1	1	1	3	2	3
Check cable laying and if applicable cable chain	1	1	2	1	1	1	1	1	1	3	2	3
Remove dirt from the photocell optics	5	5	5	5	5	5	5	5	5	5	5	5
Check condition, fixing and lengthwise extension of the ropes and belts		1		1	1	1	1	1	1	3	2	2
Check the condition and function of the counterweight system		1		1	1	1	1	1	1	3	2	2
Check condition and installation of the springs and the spring mounting, if applicable check for easy movement of the axial bearing		1			1		1		1			
Doors with ripped belt / rope or broken springs are to be immediately taken out of service		Yes		Yes	Yes	Yes	Yes	Yes	Yes			
Check leaf tensioning system		1			1	1	1					
Check for wear & tear (s) on guiding rails, especially in the area of funnel (s< 4mm)								1		3	2	3
Renew belt/s		8					8	8		15	16	16
Renew spring/s		10			9		13		9	15	16	16
Renew ropes		10		8	7		8					
Renew ropes and pulleys for counterbalance				9		11	12		14			
Renew ropes or rubber cable for curtain tension						8	8			9		
Renew flat belt***										9	9	9
Grease plastic guidings								2				

Inspection	RR300,RR200	RR300 Plus	RR300 Clean	RR300 Food	RR355	RR392	RR450	RR500	RR600	RR3000,R,Vision	RR3000 XXL	RR3000 ISO
Control box and additional attachments (e.g. impulse activator)												
Check for full number/completeness of circuit diagrams	4	4	4	4	4	4	4	4	4	4	4	4
Check main switch/CEE-plug and control box lock	4	4	4	4	4	4	4	4	4	4	4	4
Check the fixing	4	4	4	4	4	4	4	4	4	4	4	4
Electrical components												
Visual inspection for mechanical damage	1	1	2	1	1	1	1	1	1	3	2	2
Check potential balance	4	4	4	4	4	4	4	4	4	4	4	4
Safety equipment and impulse activators												
Measure self-opening height after switching off the door and document in the inspection logbook (minimum self-opening height: 2m)		1*										
Check self-test function by switching the key switch or switching the door On/Off		1*										
Function												
Check function of safety edge, photocell, additional safeties and actuators	1	1	1	1	1	1	1	1	1	1	1	1
Check function and condition of the torque arm during operation	1	1	2	1	1	1	1	1	1	3	2	2
Assess possibility of emergency opening	1	1	2	1	1	1	1	1	1	3	2	2
Check opening and closing position of the bottom profile	1	1	2	1	1	1	1	1	1	3	2	2
Check winding performance of door curtain and ropes/belts	1	1	2	1	1	1	1	1	1	3	2	2
Check function of belt and spring break switch		1*										
Check function of magnetic switch at 2m door height		1*										
Check radar motion detectors for mechanical damage and that they function properly (required detection range: whole door width, up to 1.5m in front of the door)		1*										
Check main closing edge according to EN 13241-1 respectively local standards	4	4	4	4	4	4	4	4	4	4	4	4

C. Appendix Cleaning reference RR300 Clean

The door could be cleaned with commercially available cleaning agents and disinfectant for clean room applications.

CAUTION



Risk of material damage

Improper cleaning of the door may cause corrosion and discoloration of the surface.

⇨ Do not use chlorous cleaning agents and disinfectant.

CAUTION



Risk of injury

Door movements during cleaning could cause injuries.

⇨ Set the main switch (power switch) to “Off” or “0” or disconnect the CEE plug, and secure the switch or plug against accidental re-activation or reconnection.



The maintenance and inspection work must only be performed by manufacturer trained/ authorized personnel.

The basic cleaning may only be performed by trained personnel. We recommend this to perform as part of the inspection.

Always follow the safety instructions mentioned in the user manual.

Maintenance cleaning
Clean outer surfaces with cleaning agent and disinfectant
Basic cleaning (during inspection)
Open side frames and clean side frame guides and trailing chain for cable
Clean lintel profile
Open top roll and motor cover and clean this area
Mount everything in reverse order

Periods need to be adjusted according to the cleaning concept and production relevance.

D. Appendix RR300 Food

Special notes for the RR300 Food are summarised in this annex.

D 1. Notes for the application in the food industry

The following notes are applicable to the installation and operation of RR300 Food in food processing plants. A good cleanability and integration according the hygiene examination (DGUV 11206) in the operational HACCP (Hazard Analysis and Critical Control Points) concept is possible when observing the following points. An evaluation and consideration for the respective operational requirements must be done individually.

- Door application only as interior door. The rooms are according EU regulation 852/2004 appendix II/
- Install the control box and additional parts (e.g. actuators) with a tilt angle of 3°. The distance of the control box should be min. 50 mm (2").
- Cables are fed into the control box from below through cable glands.
- Use open cable duct of stainless steel for wiring outside of the door. Observe the cleanability behind cable ducts and at fixing points.
- Install the door on the warm side when there are different temperature levels. Thereby the condensation is prevented or reduced.
- Only packaged food or food in covered containers should be moved through the door.
- Use hexagon head screws for fixing the door, the control box and additional parts.
- Seal the door to the wall. Horizontal sealings need a radius of min. 3 mm or a downgrade.
- The used additional materials (screws, cables, cable ducts, sealing compound, etc.) must be suitable for the condition of use.
- The door should be equipped the easy accessible side covers and a motor and top roll cover.

D 2. Cleaning reference RR300 Food

CAUTION

Risk of material damage

Improper cleaning of the door may cause corrosion and discoloration of the surface.

⇨ Do not use chlorous cleaning agents and disinfectant.

The maintenance and inspection work must only be performed by manufacturer trained/ authorized personnel.

Daily routine and basic cleaning must only be performed by briefed personnel.

Always follow the safety instructions mentioned in the user manual.

CAUTION



Risk of personal injury and material damage

Door movements during cleaning could cause injuries or material damage.

⇨ Set the main switch (power switch) to “Off” or “0” or disconnect the CEE plug, and secure the switch or plug against accidental re-activation or reconnection.

The door can be treated with commercial cleaners and disinfectants for the foodstuffs industry. Recommendations for cleaning agents are listed in D 2.2.

Surface cleaning

1. Close the door completely
2. Clean outer surfaces with cleaning agent and disinfectant

Daily routine maintenance cleaning

1. Close the door completely
2. Clean outer surfaces with cleaning agent and disinfectant
3. Clean side frames (chapter D 2.1.2)
4. Clean top roll cover (chapter D 2.1.3)
5. Clean rope for counterweight system (chapter D 2.1.4)
6. Carry out test run

Basic cleaning

Open the motor cover and clean this area for the basic cleaning additionally to the above listed work (chapter D 2.1.5)



The pollution of the door by cleaning the environment should be avoided. The order of the cleaning steps and the art of cleaning can avoid that a purified door is contaminated again by cleaning the periphery. If the door is again contaminated by cleaning the periphery, it must be once again be cleaned in the appropriate places (inside).



Periods need to be adjusted according to the cleaning respectively HACCP concept and production relevance.

D 2.1. Cleaning recommendations for special areas

D 2.1.1. Cleaing control cabinet

- It may get no splash water in the control cabinet.
- Do not use a water hose or pressure washer.
- Even when main switch is turned off components within the control cabinet can be electrically live!

1. Set the main switch (power switch) to “Off” or “0” or disconnect the CEE plug, and secure the switch or plug against accidental re-activation or reconnection.
2. Open the control cabinet door to clean the drainage channel on the control cabinet and the rubber seal in the control panel door.
3. Clean with a clean, damp cloth.
4. Immediately close to the control cabinet door after cleaning.

D 2.1.2. Cleaning side frame

1. Open the door up to a height of about 1m.
2. Set the main switch (power switch) to “Off” or “0” or disconnect the CEE plug, and secure the switch or plug against accidental re-activation or reconnection.

CAUTION

Risk of personal injury and material damage

Door movements during cleaning could cause injuries or material damage.

⇨ Set the main switch (power switch) to “Off” or “0” or disconnect the CEE plug, and secure the switch or plug against accidental re-activation or reconnection.

3. Open side frame:
   Loosen the lock screw which is located in about 1m height at the front of the side frame (1 turn). Move the lock screw a few mm to the side. Pull the locking lever and push upwards. Then swing open the side frame cover.

- Use a water hose for cleaning.
- Do not use high pressure cleaning machine.

4. Check cleanliness of side frames, especially of the cable chain and clean if required. A clean, soft brush with plastic bristles can be used to support the cleaning.
5. After cleaning you need to check whether all links of the cable chain are engaged.
6. Close the side frame covers and secure with the lock screw.
7. Switch on main power.
8. Carry out test run.

D 2.1.3. Cleaning inside top roll cover

1. Open the door up to a height of about 1m.
2. Set the main switch (power switch) to “Off” or “0” or disconnect the CEE plug, and secure the switch or plug against accidental re-activation or reconnection.

CAUTION



Risk of personal injury and material damage

Door movements during cleaning could cause injuries or material damage.

⇨ Set the main switch (power switch) to “Off” or “0” or disconnect the CEE plug, and secure the switch or plug against accidental re-activation or reconnection.

3. Open top roll cover:
   Loosen both screws at the front side of the top roll cover. Do not remove the screws from the top roll cover. Pull the top roll cover forward. This swings the top roll cover forward and upward and remains open in the topmost position.



- Use a water hose for cleaning.
- Do not use high pressure cleaning machine.

4. Check cleanliness inside the top roll cover and clean if required.
5. Check cleanliness of the cable channel, which connects the side frames, and the gap between cable channel and wall and clean if required.
   A clean, soft brush with plastic bristles can be used to support the cleaning.
6. Close the top roll cover and secure with the lock screw.
7. Switch on main power.
8. Carry out test run.

D 2.1.4. Cleaning the rope for the counterweight system (option)

1. Open the door completely.
2. Set the main switch (power switch) to “Off” or “0” or disconnect the CEE plug, and secure the switch or plug against accidental re-activation or reconnection.

CAUTION



Risk of personal injury and material damage

Door movements during cleaning could cause injuries or material damage.

⇨ Set the main switch (power switch) to “Off” or “0” or disconnect the CEE plug, and secure the switch or plug against accidental re-activation or reconnection.

3. Open side frame:
   Loosen the lock screw which is located in about 1m height at the front of the side frame (1 turn). Move the lock screw a few mm to the side. Pull the locking lever and push upwards. Then swing open the side frame cover.
4. Open top roll cover:
   Loosen both screws at the front side of the top roll cover. Do not remove the screws from the top roll cover. Pull the top roll cover forward. This swings the top roll cover forward and upward and remains open in the topmost position.

- Use a water hose for cleaning.
- Do not use high pressure cleaning machine.

5. Check cleanliness of the rope and the rope discs and clean if required. A clean, damp cloth can be used to support the cleaning.
6. Close the top roll cover and secure with the lock screw.
7. Switch on main power.
8. Carry out test run.

D 2.1.5. Cleaning inside the motor cover

1. Set the main switch (power switch) to “Off” or “0” or disconnect the CEE plug, and secure the switch or plug against accidental re-activation or reconnection.

CAUTION

Risk of personal injury and material damage

Door movements during cleaning could cause injuries or material damage.

⇨ Set the main switch (power switch) to “Off” or “0” or disconnect the CEE plug, and secure the switch or plug against accidental re-activation or reconnection.

2. Remove motor cover:
   Remove the screw on the bottom and two screws on the outer side of the motor cover. Then pull the motor cover forward.

- Water and a clean, damp cloth can be used for the cleaning.
- Do not use high pressure cleaning machine.

3. Check cleanliness of the drive unit, the cables and the attaching parts and clean if required. A clean, soft brush with plastic bristles can be used to support the cleaning.
4. Push the motor cover back and screw tightly.
5. Switch on main power.

D 2.2. Recommendation cleaning agents RR300 Food

This recommendation was compiled in co-operation with the companies Tana chemistry GmbH (T) (www.tana.de) and Johannes Kiehl KG (K) (www.kiehl-group.com). These companies made extensive studies of the resistance of the door materials.

Recommended cleaning agents are listed below.

daily routine maintenance cleaning		**Basic cleaning / Intensive cleaining**	
Method	Product	Method	Product
alkaline surface cleaning	Kraftreiniger (T) Torvan Konzentrat-FD (K)	Alkaline cleaning	Nowa FLA 710 S (T) Nowa Tanin (T) Grasset (K)
Alkaline spot-cleaning	Sprinter Power (T) Sprinter Grease (T)		
Alkaline disinfectant	Apesin Combi DR (T) Blutoxol (K)	Alkaline disinfectant	Apesin Combi DR (T)
Acidly disinfectant	Apesin KDR Food (T)	Acidly disinfectant	Apesin KDR Food (T)



The security council impacts and application reference on the bundles, the product information as well as the safety data sheet are to be considered.

E. Appendix RR300 Plus F+R:

E 1. Radar motion detector RR300 Plus F+R:

The best possible radar sensor settings are determined for various door heights and widths for simple putting into service of the radar sensor and for ensuring its function.



Despite the given reference values, the function of the radar should be checked again on site, as the radar field depends on various parameters and may therefore perform differently on site (required detection range: whole door width, up to 1.5m in front of the door). Note and follow the operating instructions of the radar manufacturer!

E 1.1. Reference settings for D4904R0605

Parameter		Pre-setting	Radar: Field size	Radar: Output	Radar: Immunity	Radar: Direction	Radar: Hold time	Angle setting	Radar antenna (narrow/wide)
Door height	Door width								
2.3 m	1 m	1	3-4	5	2	2	0	Adjust during installation. Start with the steepest angle setting and if necessary change so that the entire door width is covered op to 1,5m in front of the door.	narrow
	1.5 m	1	5-6	5	2	2	0		narrow
	2 m	1	7	5	2	2	0		narrow
	2.5 m	1	8-9	5	2	2	0		narrow
	3 m	1	8	5	2	2	0		wide
	3.5 m	1	9	5	2	2	0		wide
	4 m	1	9	5	2	2	0		wide
2.5 m	1 m	1	5-6	5	2	2	0		narrow
	1.5 m	1	6	5	2	2	0		narrow
	2 m	1	7	5	2	2	0		narrow
	2.5 m	1	9	5	2	2	0		narrow
	3 m	1	8	5	2	2	0		wide
	3.5 m	1	8-9	5	2	2	0		wide
	4 m	1	9	5	2	2	0		wide
3 m	1 m	1	6	5	2	2	0		narrow
	1.5 m	1	6-7	5	2	2	0		narrow
	2 m	1	7	5	2	2	0		narrow
	2.5 m	1	8	5	2	2	0		narrow
	3 m	1	9	5	2	2	0		narrow
	3.5 m	1	8	5	2	2	0		wide
	4 m	1	9	5	2	2	0		wide
3.5 m	1 m	1	7-8	5	2	2	0		narrow
	1.5 m	1	8	5	2	2	0		narrow
	2 m	1	9	5	2	2	0		narrow
	2.5 m	1	8	5	2	2	0		wide
	3 m	1	8-9	5	2	2	0		wide
	3.5 m	1	9	5	2	2	0		wide
	4 m	1	9	5	2	2	0		wide
4 m	1 m	1	9	5	2	2	0		narrow
	1.5 m	1	7-8	5	2	2	0		wide
	2 m	1	8	5	2	2	0		wide
	2.5 m	1	8	5	2	2	0		wide
	3 m	1	8-9	5	2	2	0		wide
	3.5 m	1	9	5	2	2	0		wide
	4 m	1	9	5	2	2	0		wide

Note and follow the operating instructions of the radar manufacturer!

E 2. Installation of the additional F+R components

The radar motion detector is installed in the factory in the top roll cover of doors with H<= 3700mm. In doors with H> 3700mm a separate bracket is supplied for the radar motion detector; this must be installed in the middle above the door. The maximum allowable installation height for the radar motion detector is 4000mm. Note the opening height of the door!

The solenoid actuator is installed in the factory in the side frame, on the drive side.

When installing the bottom profile the bended sheet metal plate must be mounted on the drive side as shown in Figure 1 and Figure 2. The metal plate is screwed onto the H-section.

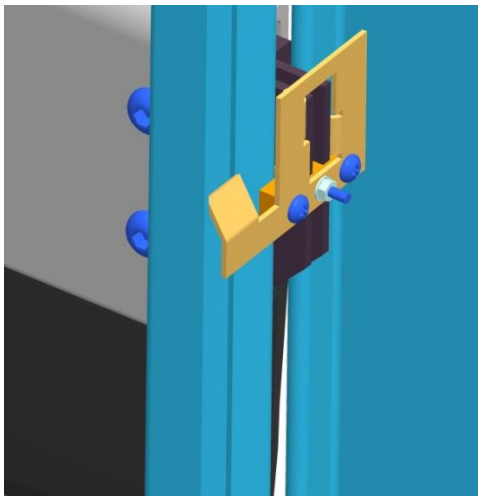


Figure 1 Display of drive side, right-hand side

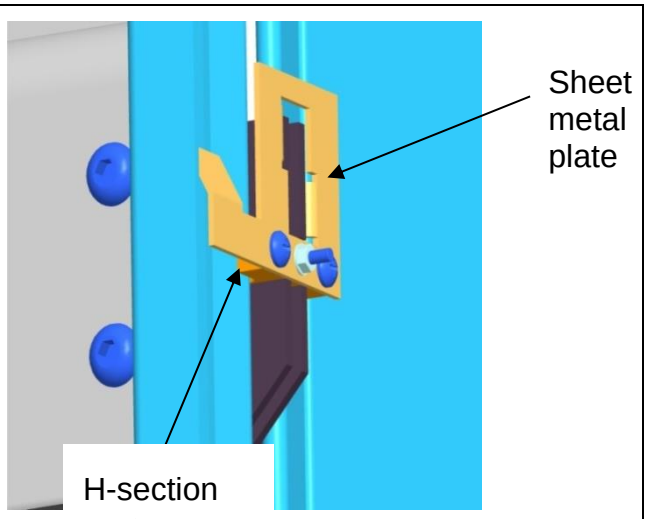


Figure 2

Hang the main spring in the spring mounting at the bottom in the side frame. The S-hook must then be hung in the spring and the eyelet of the monitoring switch, see Figure 3. To do this the fixing nuts and screws of the monitoring switch must be undone

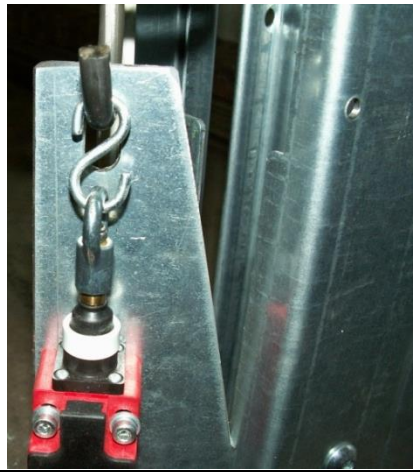


Figure 3



Figure 4

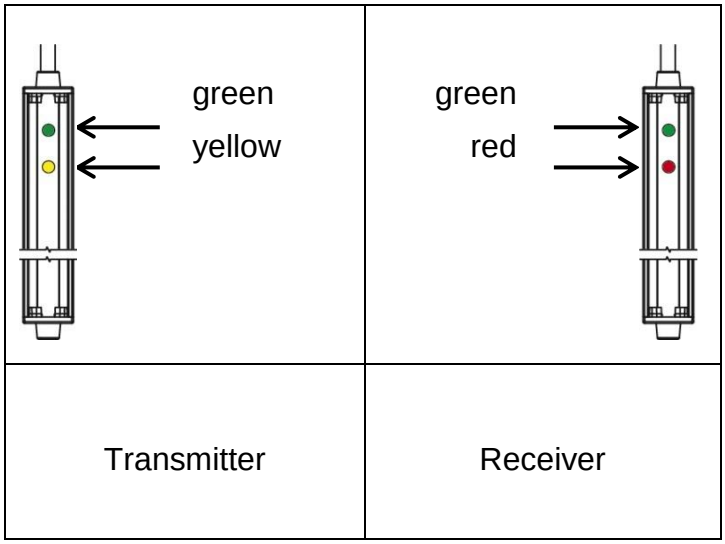
F. Appendix manual safety light grid

The safety light grid is a TÜV approved contactless working safety device. It can be mounted directly in the closing plane of the door as a stand-alone safety. The transmitter can be adjusted for door widths of 1.6 to 4m and 4 to 10m by reversing the polarity of the operating voltage. Doors with pluggable M8 junction box use an adapter cable for the range below 4m. This cable will be plugged direct to the transmitter in the sideframe and change the polarity of the power supply.

The fitting team has a detailed installation manual. In case of questions, contact ASSA ABLOY Entrance Systems.

During mounting, installation and commissioning, it must be ensured that the light grid cannot be influenced by other photocells or sources of infrared light.





F 1. Operating mode

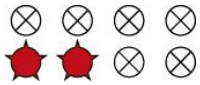
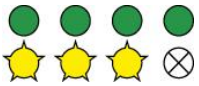
The light grid has several status LEDs, which indicate the respective operating mode.

Transmitter	
Preset door width 1.6 to 4m	
Preset door width 4 to 10m	
Test (LEDs flashing alternately)	

Receiver	
Free detection zone	
Interrupted detection zone	
Test (LEDs flashing alternately)	

F 2. Error mode

The LIGI has an internal error diagnosis function which indicates errors by means of an LED code depending on the type of error. In the event of an error, the LIGI switches to safe mode and the door can then only be operated in "hold-to-run" mode.

	Transmitter	Error mode	Receiver	
No supply voltage				Check supply voltage
Receiver polarity reversed				Check receiver operating voltage
Short at output		Red LED flashes 2x, long pause		Check output cable, overload, wrongly connected, cable defective, output on light grid defective
Error in sync cable		Yellow LED flashes 3x, long pause		Check cable, may only be connected between transmitter and receiver
Internal device error		All LEDs flashing		Light grid must be replaced